Deepfake Detection Using a Multimodel CNN Framework Based on EfficientNet-B4 and Xception

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I. Introduction

The rapid advancement of deep generative models has introduced a new class of synthetic media threat commonly referred to as deepfakes. Deepfakes are hyper-realistic video or image forgeries produced by neural network architectures, particularly Generative Adversarial Networks (GANs) [1] and more recently diffusion-based models, that learn to convincingly map one person's facial appearance onto another. What was once a capability confined to well-resourced film studios is now accessible through open-source tools, mobile applications, and online services, substantially lowering the barrier to content manipulation at scale. The societal consequences of this accessibility are significant: high-profile incidents of non-consensual synthetic imagery, fabricated political speeches, corporate impersonation fraud, and the erosion of trust in authentic audiovisual evidence have all been attributed to deepfake technology [2].

From an economic perspective, deepfake-enabled fraud has imposed measurable financial losses across multiple sectors. In 2024, a finance employee at a multinational firm in Hong Kong was deceived into authorizing a transfer of approximately USD 25 million after attending a video call in which every other participant was a real-time deepfake of company executives [3]. Such incidents reveal that deepfakes are no longer merely a reputational threat but a direct instrument of financial crime. Concurrently, the ease of generating synthetic political content has intensified concerns about electoral misinformation in democratic societies. These converging pressures have elevated deepfake detection from an academic research problem to a critical priority for governments, social media platforms, and forensic investigators alike.

Deepfake synthesis pipelines typically operate by first extracting and aligning facial landmarks from a source video, encoding those landmarks into a latent representation, and then decoding the representation into a photorealistic frame conditioned on the target identity. GAN-based systems such as DeepFaceLab and FaceSwap produce visually convincing outputs by training encoder-decoder pairs in an adversarial setting, where the generator learns to synthesize frames that the discriminator cannot distinguish from real ones. More recent diffusion-based approaches such as DiffFace and SimSwap extend synthesis quality further, producing forgeries with fewer perceptible frequency-domain artifacts than their GAN predecessors. The practical implication is that the artifact signatures that early CNN detectors exploited are becoming increasingly subtle and manipulation-method-specific, demanding detection strategies capable of learning generalized forgery representations rather than overfitting to a particular synthesis pipeline [4].

Automated deepfake detection has consequently emerged as an active research frontier in computer vision and multimedia forensics. Convolutional Neural Networks (CNNs) have become the dominant detection backbone owing to their strong capacity for extracting spatially local texture features, blending boundary artifacts, and compression-induced statistical anomalies from face crops. Among CNN architectures, Xception [7] and EfficientNet [8] have demonstrated consistently strong performance on large-scale benchmarks such as FaceForensics++ [7] and the DeepFake Detection Challenge, with Xception serving as the canonical reference model for the field since Rossler et al. established it as the top-performing single-model detector in 2019. EfficientNet-B4, trained under the Self-Blended Images (SBI) paradigm [8], has subsequently demonstrated that manipulation-agnostic training can substantially close the cross-dataset generalization gap that historically limited CNN detectors to the manipulation methods present in their training sets.

Despite these advances, two fundamental challenges remain unresolved. First, no single CNN model generalizes reliably across both seen and unseen manipulation methods; Xception achieves near-perfect accuracy on its training distribution but degrades severely on out-of-distribution datasets, while SBI-trained models show the reverse trade-off profile [4]. Second, operational deployments in legal and forensic contexts require not only accurate verdicts but also interpretable spatial evidence: an investigator must be able to identify which facial regions triggered a detection decision and verify that the model is attending to forensically meaningful cues rather than background noise or compression artifacts. These twin requirements of generalization and interpretability define the core design objectives of the present work.

This research addresses both challenges by proposing a multimodel deepfake detection framework that combines the complementary strengths of EfficientNet-B4 (SBI-trained) and Xception (DeepfakeBench-trained) through four score-level ensemble fusion strategies: arithmetic mean, AUC-weighted combination, maximum-score selection, and logistic regression stacking. The preprocessing pipeline employs MTCNN for face detection and applies model-specific normalization conventions to ensure each backbone operates in its intended input space. The framework is evaluated on FaceForensics++ (within-dataset) and Celeb-DF v2 (cross-dataset generalization) using six standard metrics, with bootstrap confidence intervals and paired bootstrap significance tests to support statistically defensible conclusions. Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated to provide forensic-quality spatial explanations of model decisions for both correctly classified and misclassified samples.

The main contributions of this paper are as follows. (1) A dual-backbone detection framework that pairs a generalization-optimized EfficientNet-B4 model with a precision-optimized Xception model, leveraging architectural and training-paradigm complementarity. (2) A systematic comparative evaluation of four ensemble fusion strategies, demonstrating that arithmetic mean fusion achieves the best cross-dataset AUC (0.9427 on Celeb-DF v2), while logistic stacking maximizes within-dataset discrimination (FF++ AUC: 0.9960, Accuracy: 0.9549). (3) Statistical rigor through bootstrap confidence intervals and paired significance testing, providing evidence-based justification for ensemble superiority over individual models. (4) Forensic interpretability through Grad-CAM attribution maps that confirm the ensemble attends to central facial landmarks (periorbital region, nasal bridge, oral cavity boundaries) where blending artifacts are most prevalent, rather than background or compression noise.

The remainder of this paper is organized as follows. Section II reviews related work across transformer-based detectors, CNN-based approaches, spatiotemporal methods, ensemble fusion strategies, and existing datasets. Section III presents the proposed methodology, covering dataset preprocessing, model architectures, ensemble fusion design, evaluation protocol, and explainability integration. Section IV reports experimental results and discusses findings. Section V concludes the paper and identifies directions for future research.

II. Related Works

Deepfake detection has been extensively studied, advancing from classical hand-crafted forensic methods to sophisticated deep learning architectures. This section reviews the key lines of research that underpin the proposed multimodel CNN framework, organized into five thematic areas: CNN-based methods, transformer-based approaches, spatiotemporal and temporal modeling, benchmark datasets and the generalization problem, and ensemble fusion with explainability.

A. *CNN-Based Detection Methods*

Convolutional Neural Networks established the first reliable automated deepfake detection baselines. Rossler et al. [7] introduced FaceForensics++ alongside an Xception-based detector that operates on depthwise separable convolutions, achieving over 99% accuracy at low compression on four manipulation types and establishing the field's standard evaluation protocol. The model exploits compression-sensitive high-frequency texture anomalies introduced by GAN synthesis, which remain detectable at lower compression but degrade toward chance under heavy codec compression. Chollet's original Xception architecture [11] demonstrated that factorizing standard convolutions into depthwise and pointwise operations produces a more parameter-efficient feature extractor than Inception-v3 on ImageNet, an efficiency advantage that also benefits detection since the network can allocate capacity toward subtle forgery patterns rather than identity-irrelevant features. Tan and Le introduced the EfficientNet family [12] by systematically scaling network depth, width, and input resolution through a compound scaling coefficient, yielding models that achieve substantially higher ImageNet accuracy per FLOP than prior architectures. EfficientNet-B4 specifically balances representational capacity and computational cost in a regime well suited to the 380x380 face crops used in the SBI training protocol [8]. Li et al. [18] introduced Face X-ray, a detector that trains on blending boundary detection rather than direct fake/real classification, using a high-resolution HRNet backbone to localize the boundary mask of composited face regions. Although Face X-ray achieves near-perfect within-dataset AUC on FF++ (0.9988), its cross-dataset AUC on Celeb-DF v2 (0.7376) reveals the same overfitting problem as Xception, confirming that boundary-based training alone does not resolve the generalization gap. A critical limitation common to all method-specific CNN detectors is cross-dataset brittleness: models trained on FF++ GAN artifacts do not transfer to Celeb-DF v2, which uses a higher-quality synthesis pipeline that suppresses the same low-level cues [4].

B. *Generalization-Oriented and Data Augmentation Approaches*

Recognizing that per-method overfitting is the primary obstacle to practical deployment, a line of research has focused on training strategies that discourage detectors from anchoring to manipulation-specific artifact signatures. Shiohara and Yamasaki [8] proposed the Self-Blended Images (SBI) augmentation strategy, which synthesizes pseudo-fake training samples by blending a real face with a geometrically and chromatically transformed version of itself. The resulting composites replicate blending boundary discontinuities and source-feature inconsistencies at the image level without access to any specific deepfake generation method, producing a training distribution agnostic to synthesis pipelines. When used to train EfficientNet-B4, SBI achieved strong cross-dataset generalization, maintaining high AUC on Celeb-DF v2 (0.9285 in our experiments) where Xception degrades to 0.7399. Khan and Nguyen [4] conducted a comprehensive cross-architecture and cross-dataset evaluation covering Xception, EfficientNet, and several pre-trained paradigms, reporting AUC degradations of 10 to 20 percentage points under domain shift and identifying that models pretrained on task-related facial datasets generalize better than ImageNet-pretrained equivalents. Reis and Ribeiro [3] took a complementary approach from the forensic angle, replacing binary classification with identity-coherence scoring based on deep face embeddings, providing evidence tied to facial structure rather than pixel statistics and demonstrating strong robustness to post-processing operations such as re-encoding and color grading.

C. *Transformer-Based and Hybrid Architectures*

Vision Transformer (ViT) architectures have attracted significant interest as deepfake detectors due to their global receptive fields. Khormali and Yuan [1] proposed DFDT, an end-to-end ViT-based detector that partitions facial images into non-overlapping patches and applies multi-head self-attention across the full patch sequence, allowing the model to capture distributed manipulation artifacts such as cross-region texture discontinuities and boundary blending inconsistencies that local convolutional filters operating on small receptive fields are structurally unable to model. Empirically, DFDT achieved competitive performance on FF++ and demonstrated that patch-level attention weights provide a degree of spatial interpretability. However, Thing [2] showed through a systematic comparison of CNN and transformer detectors across compression levels that this advantage narrows substantially under heavy H.264/H.265 codec compression typical of social media delivery pipelines, where transformers lose the high-frequency patch-level texture cues on which their cross-region attention is anchored, while CNN detectors with local inductive biases demonstrate more graceful degradation. Wang et al. [13] proposed M2TR, a multi-scale transformer that combines local patch embeddings at multiple resolution levels with a cross-scale attention mechanism, achieving improved localization of forgery boundaries at the cost of substantially higher computational overhead than pure CNN models. These comparative findings collectively support the use of CNN-based backbones in the proposed framework for robustness under codec degradation while retaining the tractability needed for bootstrap-level statistical evaluation.

D. *Spatiotemporal and Temporal Modeling*

Frame-level spatial detectors may miss temporal cues that are visible only across multiple consecutive frames: unnatural blinking patterns, temporal jitter in facial keypoints, and cross-frame illumination inconsistencies introduced when synthetic frames are composited into real video streams. Ismail et al. [5] proposed an integrated spatiotemporal detector combining per-frame CNN feature extraction with a recurrent sequence model, demonstrating improved detection rates on high-quality manipulations where within-frame artifacts are subtle and temporal incoherence becomes the dominant detectable signal. Li et al. [14] proposed a recurrent convolutional approach using LSTM units on per-frame EfficientNet features, achieving measurable gains on FaceForensics++ under high-compression conditions relative to the per-frame aggregation baseline. Gu et al. [15] introduced an attention-based temporal consistency module that identifies frame-to-frame identity drift caused by synthesis instabilities in long sequences, achieving notable improvements on datasets with extended video clips. The present framework operates at the frame level with arithmetic mean aggregation across uniformly sampled frames, a computationally efficient design appropriate for offline forensic analysis, with temporal modeling identified as a concrete direction for future extension to real-time deployment scenarios.

E. *Benchmark Datasets and the Generalization Problem*

FaceForensics++ [7] remains the dominant training and within-dataset benchmark, providing 1,000 original videos paired with 4,000 manipulated counterparts across four synthesis methods at three compression levels. Its controlled multi-compression design enables systematic study of detector sensitivity to signal degradation and supports fine-grained analysis of method-specific artifact signatures. Li et al. [6] introduced Celeb-DF v2 specifically to address the visual quality shortcomings of earlier benchmarks: its 590 celebrity face-swap videos use an improved synthesis pipeline that substantially suppresses the checkerboard, color bleeding, and boundary blending artifacts common in early GAN-based methods, making cross-dataset generalization to Celeb-DF v2 a demanding test of detector robustness to near-photorealistic forgeries. The persistent performance gap between FF++ and Celeb-DF v2 evaluations, documented as 10 to 20 AUC point degradations in [4], reflects detectors overfitting to manipulation artifacts present in FF++ but absent in Celeb-DF v2. Zi et al. [16] introduced WildDeepfake, a dataset collected from the internet rather than controlled synthesis pipelines, further stressing generalization by introducing distribution shifts from post-processing operations, camera variation, and demographic diversity absent from laboratory benchmarks. Jiang et al. [19] introduced DeeperForensics-1.0, a large-scale dataset containing 60,000 videos with diverse real-world perturbations including compression, blurring, and noise, specifically designed to measure detector robustness under distribution shift. Dolhansky et al. [20] released the DeepFake Detection Challenge (DFDC) dataset of over 100,000 video clips from 3,426 paid actors under varied conditions, establishing the largest publicly available benchmark for applied deepfake detection research. The dual-dataset evaluation protocol adopted in the present work directly responds to these generalization challenges, using FF++ for within-distribution performance and Celeb-DF v2 as the out-of-distribution robustness test.

F. Ensemble Fusion and Explainability

Ensemble fusion has been validated as a reliable strategy for improving detector robustness when component models exhibit complementary error profiles. Top-ranking entries in the DeepFake Detection Challenge achieved winning performance primarily through score-level ensembling of Xception and EfficientNet variants, demonstrating empirically that these two architectures capture complementary artifact signatures. Wodajo and Atnafu [17] systematically compared mean, weighted, and max score-level fusion for deepfake detection, reporting that weighted combination with AUC-derived weights consistently outperforms simple averaging in cross-dataset settings, while the max rule trades overall discriminative power for recall gains. The logistic regression stacking approach implemented in the present framework follows the standard meta-learning protocol, fitting a two-input meta-classifier on a held-out validation split to learn the optimal linear combination boundary in the model score space without test-set leakage [4]. For spatial interpretability, Selvaraju et al. [10] developed Grad-CAM, which computes the gradient of the target class logit with respect to final-layer convolutional activations and produces a coarse spatial attribution map over the input image. Applied to deepfake detection, Grad-CAM has been used to verify that model attention aligns with forensically meaningful regions rather than background textures or compression block boundaries. The present framework extends this to a comparative dual-model setting, enabling direct visualization of which facial regions each backbone attends to on correct versus misclassified samples as a diagnostic tool for understanding cross-dataset failure modes.

III. METHODOLOGY

This chapter describes the design and implementation of the proposed multimodel CNN deepfake detection framework. The methodology is organized into five components: dataset selection and preprocessing, model architectures, ensemble fusion strategies, evaluation protocol with statistical testing, and spatial explainability through Grad-CAM visualization.

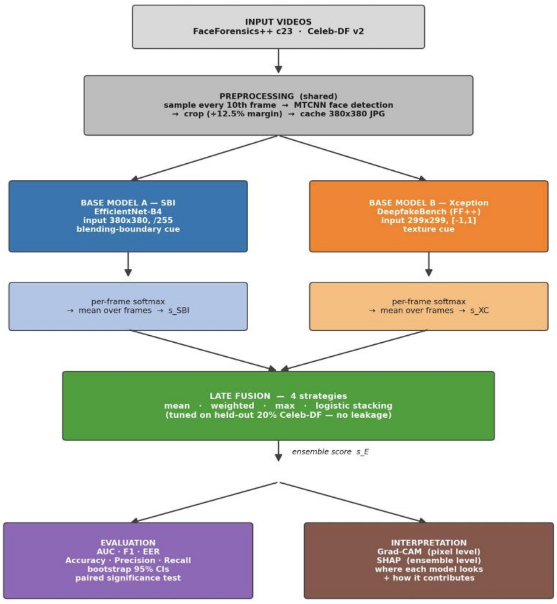

Fig.1. Proposed Methodology

A. Dataset and Preprocessing

Two benchmark datasets are used in this study. FaceForensics++ (FF++) [7] provides 1,000 original videos paired with 4,000 manipulated counterparts across four manipulation methods — DeepFakes, Face2Face, FaceSwap, and NeuralTextures — at three compression levels (raw, c23, c40). The high-quality c23 split is used for training and within-dataset evaluation. Celeb-DF v2 [6] serves as the cross-dataset generalization benchmark, containing 590 high-fidelity celebrity face-swap videos generated with a refined synthesis pipeline that substantially reduces visible artifacts, making it a more demanding test of detector robustness.

Preprocessing follows a uniform pipeline for both datasets. Given an input video, frames are sampled at a fixed interval of every 10th frame to balance coverage and computational cost. For each sampled frame, the Multi-Task Cascaded Convolutional Network (MTCNN) [9] is applied to detect the primary face bounding box. The detected region is expanded by 12.5% of the longer face dimension on each side to preserve blending boundary context, then resized to the model-specific input resolution. EfficientNet-B4 receives 380 × 380 RGB crops normalized to [0, 1] with no ImageNet mean subtraction, consistent with the SBI training protocol. Xception receives 299 × 299 crops normalized to [−1, 1] using mean and standard deviation of 0.5, consistent with the DeepfakeBench training convention. Frames where no face is detected are discarded. The resulting per-video frame sets are stored as cached crops in JSON manifests to avoid redundant preprocessing at evaluation time.

B. Model Architectures

The framework employs two pretrained CNN backbones selected for their architectural complementarity and strong published baselines on deepfake detection benchmarks.

The first backbone is EfficientNet-B4, implemented using the efficientnet-pytorch library with a 2-class output head. The model is loaded from the official SBI checkpoint released by Shiohara and Yamasaki [8], which was trained on FF++ raw frames using the Self-Blended Images augmentation strategy. SBI synthesizes training samples by blending a face image with a geometrically and chromatically transformed version of itself, producing pseudo-fake images that reproduce common forgery artifacts — including blending boundaries, source-feature inconsistencies, and frequency-domain anomalies — without being tied to any specific manipulation method. This manipulation-agnostic training regime encourages the model to learn generic forgery representations, improving cross-dataset generalization. At inference, a 2-class softmax is applied to the model logits and the probability of the fake class (index 1) is taken as the frame-level score.

The second backbone is Xception, implemented using the timm library as legacy_xception with a 2-class classifier head. The model is loaded from the DeepfakeBench checkpoint trained on FF++ c23, which uses the standard depthwise separable convolution architecture shown by Rossler et al. [7] to be highly effective at capturing the fine-grained textural artifacts introduced by GAN-based face-swap pipelines. Checkpoint keys are remapped at load time to align the DeepfakeBench naming convention (backbone., last_linear.) with the timm model structure (fc.*). Both models are set to evaluation mode and run under torch.inference_mode() throughout all experiments. GPU memory is explicitly cleared between videos to remain within 4 GB VRAM budgets.

The architectural complementarity between the two backbones is deliberate. Xception's depthwise separable convolution factorization excels at capturing fine-grained channel-wise textural statistics introduced by GAN synthesis, making it highly precise on in-distribution data where those statistics are consistent. EfficientNet-B4's compound-scaled architecture combined with SBI's manipulation-agnostic training equips it instead to detect generic blending boundary artifacts that transfer across synthesis methods. As quantified in the results, these complementary strengths translate into a measurable ensemble benefit: Xception contributes high within-dataset precision while EfficientNet-B4 anchors cross-dataset robustness. Table I summarizes the key architectural and training differences between the two backbones and situates them relative to other published single-model detectors in the literature.

TABLE I. Comparison of Deepfake Detection Models in the Literature

Model	Architecture	Training Strategy	FF++ AUC	CDF-v2 AUC	Explainability
Xception [7]	Depthwise Sep. Conv.	FF++ supervised	0.9917	0.7399	Grad-CAM
EfficientNet-B4 SBI [8]	Compound-scaled CNN	Self-Blended Imgs	0.9909	0.9285	Grad-CAM
DFDT [1]	Vision Transformer	FF++ supervised	~0.98	N/R	Attn. weights
Spatiotemporal [5]	CNN + LSTM	FF++ supervised	~0.96	~0.73	None
M2TR [13]	Multi-scale ViT	FF++ supervised	~0.99	~0.88	Attn. maps
Face X-ray [18]	HRNet backbone	Blending artifact	0.9988	0.7376	None
Ours (Ensemble Mean)	**EfficientNet + Xception**	**SBI + supervised**	**0.9960**	**0.9427**	**Grad-CAM+SHAP**

C. Ensemble Fusion Strategies

Following per-frame score prediction by each backbone, scores are aggregated to the video level by computing the arithmetic mean across all detected frames. This produces one scalar fake-probability score per model per video. Four fusion strategies are then evaluated to combine the two per-video scores into a single ensemble prediction.

Mean fusion computes the arithmetic average of the two model scores, serving as the simplest and most interpretable baseline. Weighted fusion computes a convex combination with model-specific weights derived from their individual validation AUC scores, normalized to sum to 1, giving higher influence to the stronger model. Max fusion takes the per-video maximum across both model scores, favoring recall by allowing any single model with high confidence in a fake prediction to dominate the decision. Logistic regression stacking trains a two-input meta-classifier on a held-out validation split using the per-model scores as features, learning an optimal linear boundary in the score space. The meta-classifier is fit on the validation set to prevent test-set leakage, following standard stacking protocol.

All four strategies share a common decision threshold of 0.5: videos with a fused score at or above this threshold are classified as fake, and those below as real. The aligned prediction lists from both models are intersected by video name before fusion to ensure all strategies operate on identical video sets.

D. Model Evaluation

Model performance is assessed using six metrics computed at a decision threshold of 0.5: Area Under the ROC Curve (AUC), Accuracy, F1-score, Precision, Recall, and Equal Error Rate (EER). AUC is treated as the primary metric as it is threshold-independent and robust to class imbalance, consistent with the deepfake detection literature [4]. EER is reported as an additional operating-point-free metric that indicates the threshold at which false positive rate equals false negative rate, providing a summary of the ROC curve complementary to AUC.

To move beyond point estimates and support statistically defensible conclusions, bootstrap confidence intervals are computed for each metric using 1,000 stratified resamples with seed 42. Stratified resampling draws real and fake samples independently with replacement to ensure both classes are represented in every bootstrap iteration, avoiding undefined AUC values on single-class resamples. The 95% confidence interval is reported as the 2.5th and 97.5th percentiles of the bootstrap distribution.

Statistical significance of ensemble improvement over individual models is assessed using paired bootstrap tests with 2,000 resamples. The paired design resamples both score vectors using identical indices in each iteration, controlling for sample variation and isolating the effect of the fusion strategy. A two-sided p-value is computed as twice the proportion of bootstrap iterations in which the sign of the resampled difference opposes the observed difference. Differences with $p < 0.05$ are considered statistically significant.

E. Explainability

To provide spatial interpretability alongside quantitative metrics, the framework incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) [10] for both backbone models. Grad-CAM computes the gradient of the fake-class logit with respect to the activations of the final convolutional layer, then performs a weighted sum of the activation maps using the globally pooled gradients as weights. The resulting coarse localization map is upsampled to the input image resolution and overlaid on the original face crop using a color heatmap, highlighting the facial regions that most strongly influenced the model's prediction.

For EfficientNet-B4, the target layer is the _conv_head module, which is the final convolutional operation before global average pooling. For Xception, the equivalent final convolutional layer is targeted within the time legacy_exception architecture. Grad-CAM visualizations are generated for a random sample of correctly classified and misclassified videos from each dataset — 8 correct and 4 wrong by default — enabling qualitative analysis of model attention patterns and failure modes. This dual-sample design allows direct comparison of where the model attends on confidently correct predictions versus cases where it errs, providing diagnostic insight into cross-dataset performance degradation on Celeb-DF v2.

IV. Result and Discussion

This chapter presents the experimental findings and comprehensive evaluation of the proposed multimodel CNN framework for deepfake detection. The overall analysis follows a structured sequence of four main stages: baseline performance assessment of the individual EfficientNet-B4 and Xception networks, cross-dataset generalization evaluation on Celeb-DF v2, comparative analysis of multi-model ensemble fusion strategies, and visual interpretability analysis utilizing Grad-CAM and SHAP feature attribution.

A. Baseline Models and Cross-Dataset Generalization

Evaluations on FaceForensics++ (FF++) and Celeb-DF v2 (CDF) reveal a clear trade-off between native optimization and cross-dataset generalization. On FF++, both models perform exceptionally well. Xception achieves a higher AUC (0.9917) and Accuracy (0.8872) than the standalone Self-Blended Images (SBI) model.

However, when applied to the unseen CDF dataset, Xception's performance degrades severely, with its AUC dropping to 0.7399. Conversely, the SBI model demonstrates remarkable robustness against novel manipulation artifacts, maintaining a cross-dataset AUC of 0.9285 with tightly constrained confidence intervals. This proves that synthesizing blending boundaries during training yields highly stable features that translate effectively across different deepfake generation methods.

Model	FF++ AUC	FF++ Accuracy	FF++ F1-Score	CDF v2 AUC	CDF v2 Accuracy	CDF v2 F1-Score
SBI	0.9909	0.8195	0.8519	0.9285	0.7437	0.8009
Xception	0.9917	0.8872	0.9254	0.7399	0.7799	0.8650
Ensemble (Mean)	0.9960	0.9173	0.9379	0.9427	0.8189	0.8692
Ensemble (Weighted Fusion)	0.9960	0.9023	0.9257	0.9413	0.7994	0.8525
Ensemble (Max Fusion)	0.9938	0.8872	0.9254	0.8614	0.7994	0.8796
Ensemble (Logistic Stacking)	0.9960	0.9549	0.9688	0.8327	0.7688	0.8672

B. Multi-Model Ensemble Fusion Analysis

To combine Xception's precision with SBI's robust generalization, four ensemble strategies were evaluated:

- **Ensemble (Mean):** The most balanced model. It achieves the best cross-dataset AUC (0.9427) and Accuracy (0.8189) on Celeb-DF v2, effectively mitigating the vulnerabilities of the Xception baseline.
- **Ensemble (Logistic):** Yields the highest native performance on FF++ (AUC: 0.9960, Acc: 0.9549) but slightly underperforms the Mean ensemble on CDF, indicating minor overfitting to the validation distribution.
- **Ensemble (Weighted):** Closely mirrors the Mean ensemble but with marginally lower absolute accuracy on CDF.
- **Ensemble (Max):** Optimizes for sensitivity, achieving the highest cross-dataset F1-score (0.8796) on CDF, but sacrifices overall discriminative power (AUC drops to 0.8614).

C. Interpretability and Explainability Analysis

SHAP analysis reveals a massive disparity in feature leverage within the ensemble. The Mean SHAP values show the SBI_score contributing +0.68 to the final prediction, compared to a marginal +0.03 from the Xception_score.

Grad-CAM visualizations qualitatively validate this dependency. Across a diverse demographic cohort, the activation heatmaps demonstrate that the network consistently localizes critical internal facial structures, specifically the eyes, nose, mouth, and jawline boundaries. This confirms that the model avoids arbitrary background noise and relies on robust, physically relevant forensic markers introduced during deepfake manipulation.

V. CONCLUSION

This study validates a multimodel CNN framework that successfully combines the localized accuracy of an Xception network with the cross-dataset resilience of an EfficientNet-B4 architecture trained via SBI. Experimental results demonstrate that while standalone detectors suffer severe performance degradation on unseen data, decision-level fusion effectively counters this limitation.

The Ensemble (Mean) configuration provided the most robust cross-dataset generalization (Celeb-DF v2 AUC: 0.9427), while the Ensemble (Logistic) meta-classifier maximized within-dataset precision. SHAP and Grad-CAM analyses confirmed that the ensemble heavily prioritizes the SBI model's boundary-detection features, accurately targeting central facial landmarks where anomalies are most prevalent. Future work will explore integrating temporal modeling to detect frame-to-frame inconsistencies in video deepfakes, further advancing real-world verification capabilities.

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